

GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval

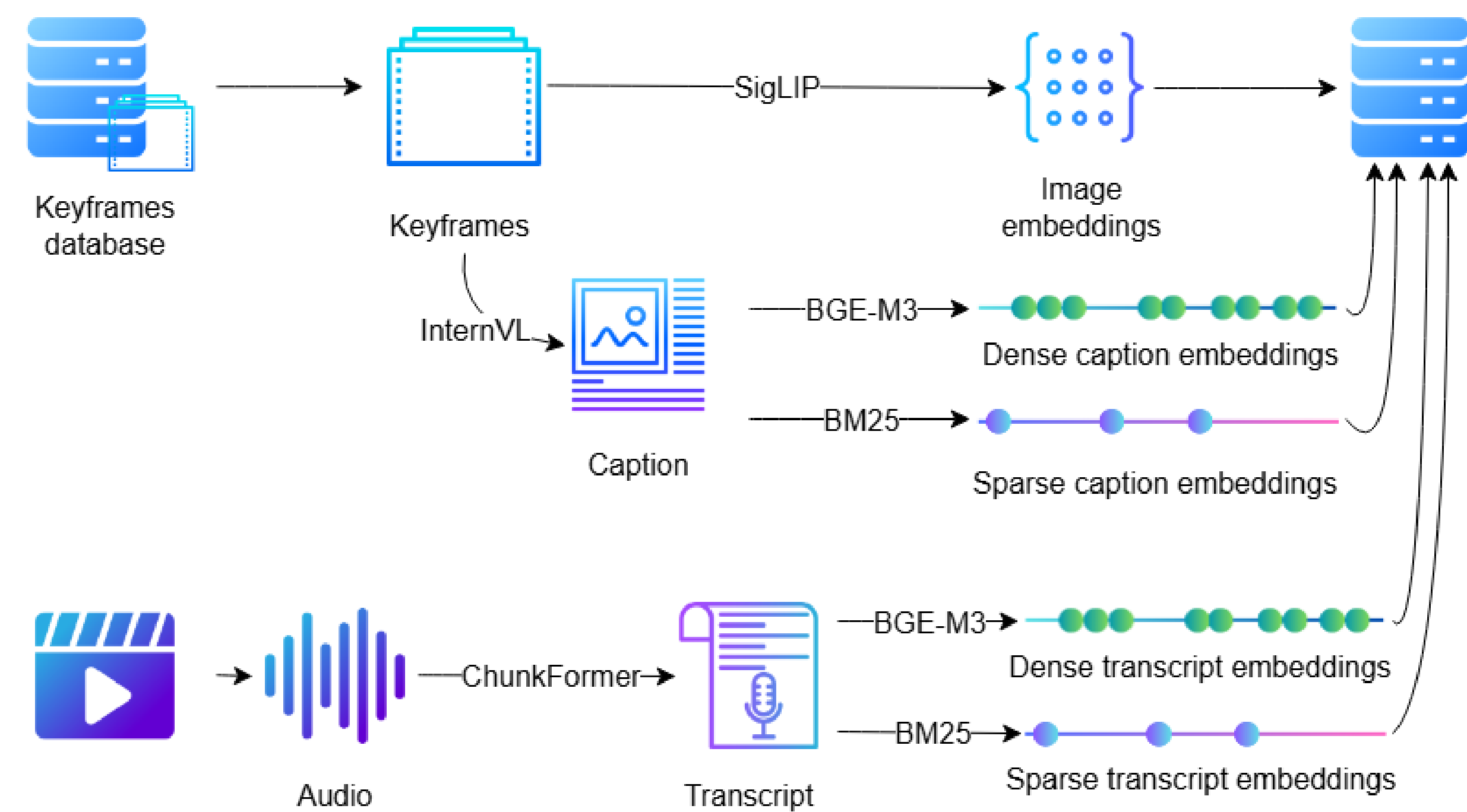
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Introduction

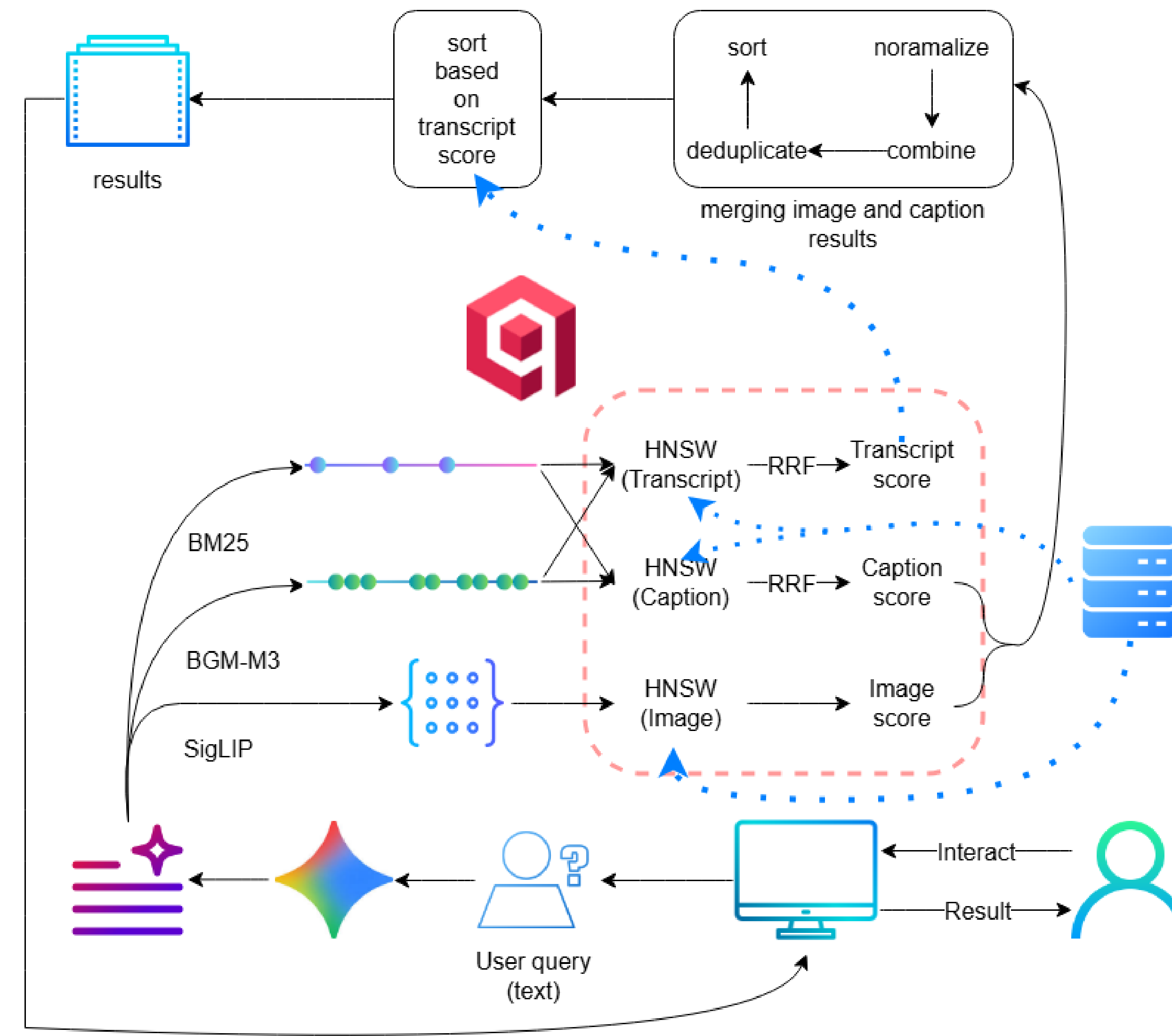
The rapid growth of large-scale video collections has created significant challenges for retrieving events that occur across varying visual, audio, and textual contexts. Traditional unimodal retrieval systems are often inadequate because a query may refer to complex visual scenes, spoken dialogue, or abstract semantic concepts that cannot be captured by a single modality. Major video retrieval challenges have increasingly demonstrated the need for systems that can jointly understand images, captions, and transcripts. To address this gap, GalaxyAssistant introduces a tri-modal retrieval framework capable of analyzing videos through visual embeddings, caption-based semantics, and audio-transcribed content, enabling richer and more accurate event-based search.

Methodology



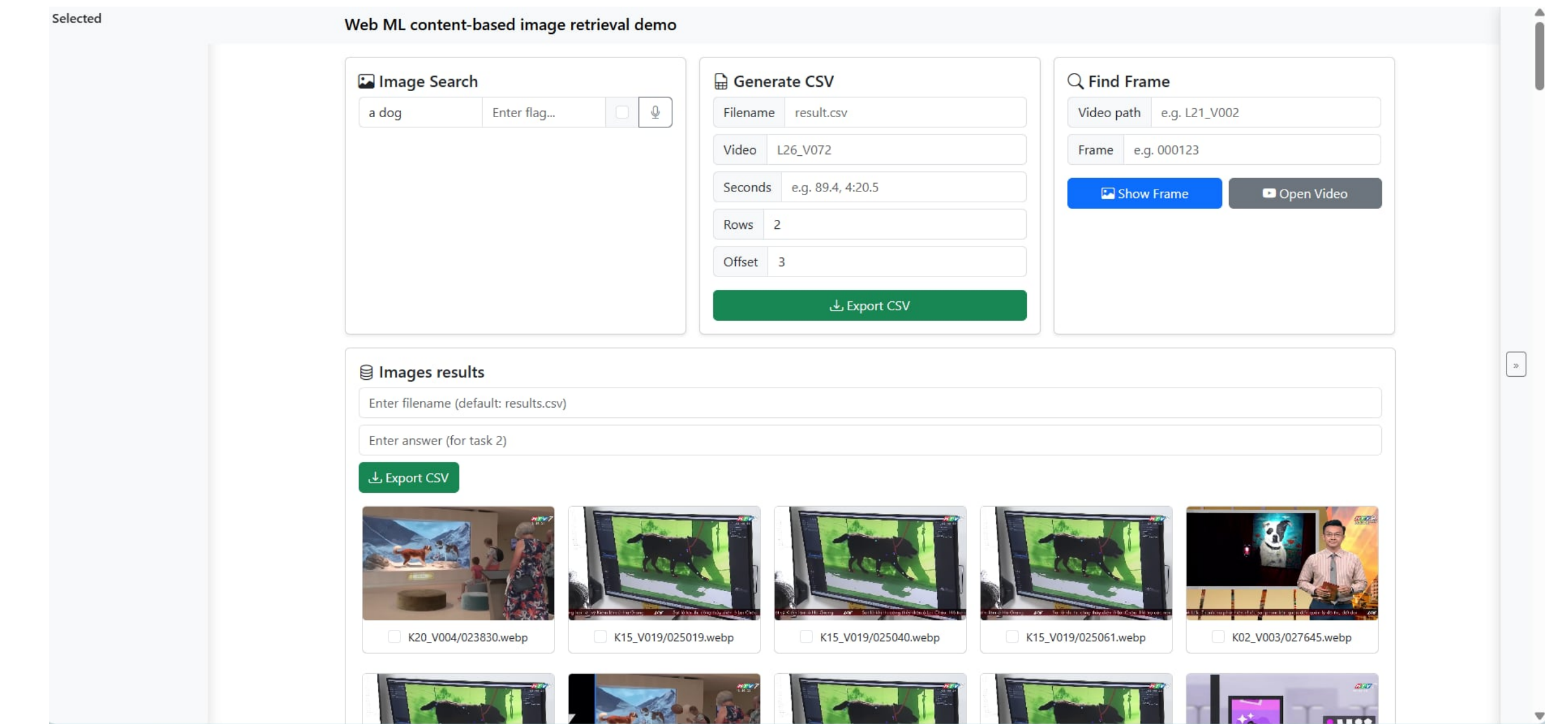
During the offline stage, videos are segmented into scenes using TransNetV2, and keyframes are extracted at representative points within each segment. These keyframes are encoded by SigLIP to produce visual embeddings. In parallel, InternVL generates captions for each keyframe, which are then transformed into both dense and sparse text embeddings using BGE-M3 and BM25. The audio track is transcribed with ChunkFormer, and the transcripts are processed through the same hybrid text embedding pipeline. All embeddings across all modalities are stored in Qdrant using HNSW indexing to support scalable approximate nearest-neighbor search.

During the online retrieval stage, the user query is enhanced with Gemini 2.5 Flash-Lite and encoded using the same tri-modal embedding scheme. Searches are run concurrently across all modalities, caption and transcript scores are fused with Reciprocal Rank Fusion, and final results are reranked together with visual scores to produce highly relevant retrieval outputs.



User Interface

The GalaxyAssistant interface is designed to provide a clean and intuitive experience that supports the full retrieval workflow. The main search panel accepts text query and displays the system's results as a grid of keyframes that users can view, inspect, and compare. By keeping the design minimal and user-centered, the interface balances functionality with clarity, enabling smooth interaction with the system's multimodal retrieval abilities.



Conclusion

Our team achieved a score of 78 in the Preliminary Round. In the Final Round, our performance was assessed as Satisfactory for TKIS, Very Good for VKIS and TRAKE, and Excellent for QA, resulting in an overall Good performance. Its multimodal fusion pipeline enables effective retrieval of events that would be difficult to detect through a single modality alone. However, the system's performance is dependent on the quality of pretrained models used for captioning, speech transcription, and visual encoding, which can introduce errors when operating on noisy or domain-specific content. Additionally, the system does not yet integrate OCR or object detection, how to handle temporal features, simple fusion techniques, limiting its ability to handle text-in-image queries or fine-grained object-based search. Despite these limitations, GalaxyAssistant offers a strong and extensible foundation for multimodal video retrieval and competitive performance in real-world challenge settings.

References

- [1] T.-L. Do, V.-T. Huynh, H.-D. Nguyen, T. Nguyen-Quang, M.-K. Tran, T.-T. Nguyen, T. V. Ninh, T.-K. Le, T. D. Ngo, D.-T. Dang-Nguyen, T.-T. Ngo, K. Schöffmann, C. Gurrin, and M.-T. Tran. Toward abstraction-level event retrieval in large video collections: Leveraging human knowledge and LLM-based reasoning in the Ho Chi Minh City AI Challenge 2025. In *Proceedings of the 14th International Symposium on Information and Communication Technology (SOICT 2025)*, CCIS, Nha Trang, Vietnam, 2025. Springer.
- [2] G. Gemini Team. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities, 2025. URL <https://arxiv.org/abs/2507.06261>.